Practical No: 07

A)Develop a PWA that works in offline mode using service Worker.

Project Name: OfflinePWA

Project Type: MVC App

Program Code

I.Filename: Index.cshtml

```
<div class="text-center">
    <h1>College Information Portal</h1>
    <h3>PWA Offline Demo</h3>
    <p>This application works without internet connection.</p>
</div>
```

II.Filename: manifest.json

```
{
  "name": "Offline PWA",
  "short_name": "OfflinePWA",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

III.Filename: service-worker.js

```
const CACHE_NAME = "pwa-cache-v1";
```

```
const urlsToCache = [
    "/",
    "/Home/Index"
];

self.addEventListener("install", event => {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache => {
                return cache.addAll(urlsToCache);
            })
    );
});

self.addEventListener("fetch", event => {
    event.respondWith(
        caches.match(event.request)
            .then(response => {
                return response || fetch(event.request);
            })
    );
});
```

IV.Filename: Layout.cshtml

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>@ViewData["Title"] - _SWOfflinePWA_</title>

    <script type="importmap"></script>

    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
    <link rel="stylesheet" href="~/_SWOfflinePWA_.styles.css" asp-append-
version="true" />
```

```
    <!-- PWA Manifest -->
    <link rel="manifest" href="/manifest.json">
</head>

<body>
    <header>
        <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-
white border-bottom box-shadow mb-3">
            <div class="container-fluid">

                <a class="navbar-brand"
                   asp-area=""
                   asp-controller="Home"
                   asp-action="Index">
                    _SWOfflinePWA_
                </a>

                <button class="navbar-toggler"
                        type="button"
                        data-bs-toggle="collapse"
                        data-bs-target=".navbar-collapse"
                        aria-controls="navbarSupportedContent"
                        aria-expanded="false"
                        aria-label="Toggle navigation">
                    <span class="navbar-toggler-icon"></span>
                </button>

                <div class="navbar-collapse collapse d-sm-inline-flex justify-
content-between">
                    <ul class="navbar-nav flex-grow-1">

                        <li class="nav-item">
                            <a class="nav-link text-dark"
                               asp-area=""
                               asp-controller="Home"
                               asp-action="Index">
                                Home
```

```
                        </a>
                    </li>

                    <li class="nav-item">
                        <a class="nav-link text-dark"
                           asp-area=""
                           asp-controller="Home"
                           asp-action="Privacy">
                            Privacy
                        </a>
                    </li>

                </ul>
            </div>
        </div>
    </nav>
</header>

<div class="container">
    <main role="main" class="pb-3">
        @RenderBody()
    </main>
</div>

<footer class="border-top footer text-muted">
    <div class="container">
        &copy; 2026 - _SWOfflinePWA_ -
        <a asp-area=""
           asp-controller="Home"
           asp-action="Privacy">
            Privacy
        </a>
    </div>
</footer>

<script src="~/lib/jquery/dist/jquery.min.js"></script>
<script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
```

```
    <script src="~/js/site.js" asp-append-version="true"></script>


    @await RenderSectionAsync("Scripts", required: false)


    <!-- Register Service Worker -->
    <script>
        if ('serviceWorker' in navigator) {
            navigator.serviceWorker.register('/service-worker.js')
                .then(() => console.log('Service Worker Registered'))
                .catch(error => console.error('Service Worker Registration
Failed:', error));
        }
    </script>
</body>
</html>
```

B)Create a Student Record Management PWA.

Project Name: StudentRecordPWA

Project Type: MVC App

Program Code

I.Filename: Index.cshtml

```
@model IEnumerable<StudentRecordPWA.Models.Students>


<h2>Student Records</h2>


<table border="1" cellpadding="10">
    <tr>
        <th>Id</th>
        <th>Name</th>
        <th>Course</th>
        <th>Age</th>
    </tr>

    @foreach (var item in Model)
    {
        <tr>
            <td>@item.Id</td>
            <td>@item.Name</td>
            <td>@item.Course</td>
            <td>@item.Age</td>
        </tr>
    }
</table>
```

II.Filename: manifest.json

```
{
  "name": "Student Record PWA",
  "short_name": "StudentPWA",
  "start_url": "/Students",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
```

```
    {
      "src": "icons/icon1.png",
      "sizes": "192x192",
      "type": "image/png"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png"
    }
  ]
}
```

III.Filename: service-wroker.js

```
const CACHE_NAME = "student-pwa-cache";


const urlsToCache = [
    "/Students"
];


self.addEventListener("install", event => {
    event.waitUntil(
        caches.open(CACHE_NAME)
            .then(cache => {
                return cache.addAll(urlsToCache);
            })
    );
});


self.addEventListener("fetch", event => {
    event.respondWith(
        caches.match(event.request)
            .then(response => {
                return response || fetch(event.request);
            })
    );
});
```

IV.Filename: Layout.cshtml

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="utf-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />


    <title>@ViewData["Title"] - StudentRecordPWA</title>


    <script type="importmap"></script>


    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />

    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />

    <link rel="stylesheet" href="~/StudentRecordPWA.styles.css" asp-append-
version="true" />


    <!-- PWA Manifest -->

    <link rel="manifest" href="/manifest.json">

</head>


<body>

    <header>

        <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-
white border-bottom box-shadow mb-3">

            <div class="container-fluid">


                <a class="navbar-brand"

                   asp-area=""

                   asp-controller="Home"

                   asp-action="Index">

                    StudentRecordPWA

                </a>


                <button class="navbar-toggler"

                        type="button"

                        data-bs-toggle="collapse"

                        data-bs-target=".navbar-collapse"

                        aria-controls="navbarSupportedContent"

                        aria-expanded="false"
```

```
                        aria-label="Toggle navigation">
                    <span class="navbar-toggler-icon"></span>
                </button>

                <div class="navbar-collapse collapse d-sm-inline-flex justify-
content-between">
                    <ul class="navbar-nav flex-grow-1">

                        <li class="nav-item">
                            <a class="nav-link text-dark"
                               asp-area=""
                               asp-controller="Home"
                               asp-action="Index">
                                Home
                            </a>
                        </li>

                        <li class="nav-item">
                            <a class="nav-link text-dark"
                               asp-area=""
                               asp-controller="Home"
                               asp-action="Privacy">
                                Privacy
                            </a>
                        </li>

                    </ul>
                </div>
            </div>
        </nav>
    </header>

    <div class="container">
        <main role="main" class="pb-3">
            @RenderBody()
        </main>
    </div>

    <footer class="border-top footer text-muted">
```

```
        <div class="container">
            &copy; 2026 - StudentRecordPWA -
            <a asp-area=""
               asp-controller="Home"
               asp-action="Privacy">
                Privacy
            </a>
        </div>
    </footer>

    <script src="~/lib/jquery/dist/jquery.min.js"></script>
    <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
    <script src="~/js/site.js" asp-append-version="true"></script>

    @await RenderSectionAsync("Scripts", required: false)

    <!-- Register Service Worker -->
    <script>
        if ('serviceWorker' in navigator) {
            navigator.serviceWorker.register('/service-worker.js')
                .then(() => console.log('Service Worker Registered'))
                .catch(error =>
                    console.error('Service Worker Registration Failed:', error)
                );
        }
    </script>
</body>
</html>
```

V.Filename: StudentsController.cs

```
using Microsoft.AspNetCore.Mvc;
using StudentRecordPWA.Models;
using System.Collections.Generic;

namespace StudentRecordPWA.Controllers
{
    public class StudentsController : Controller
    {
        public IActionResult Index()
```

```
        {
            List<Students> students = new List<Students>()
            {
                new Students
                {
                    Id = 1,
                    Name = "Rahul",
                    Course = "BSc IT",
                    Age = 20
                },

                new Students
                {
                    Id = 2,
                    Name = "Amit",
                    Course = "BCA",
                    Age = 21
                },

                new Students
                {
                    Id = 3,
                    Name = "Sneha",
                    Course = "BSc CS",
                    Age = 19
                }
            };

            return View(students);
        }
    }
}
```

V.Filename: Students.cs

```
namespace StudentRecordPWA.Models
{
    public class Students
    {
```

```
        public int Id { get; set; }


        public string Name { get; set; }


        public string Course { get; set; }


        public int Age { get; set; }
    }
}
```